



BW WATER

Technical Proposal for ADASA's Taltal Desalination Plant – Brine Recovery Project - UHPRO System

**Client Ref.: Proceso de Licitación 12803 SUMINISTRO
MÓDULO RO SEGUNDA ETAPA PARA SALMUERA -
PLANTA DESALADORA TALTAL (ex 12783)**

To: Aguas Antofagasta S.A. (ADASA)

BW Water Americas Inc.

**BWWA Ref. No: 20.24.6501.F - Rev.0
August 5, 2025 | Confidential**

4520 Oak Fair Blvd, Suite 200
Tampa, FL 33610 - USA
www.bw-water.com.com

Technical Proposal

PROPRIETARY, PATENT NOTICE & CONFIDENTIALITY AGREEMENT

This proposal and the concept of the system are the intellectual property of BW Water and its Affiliates and are subject to legal copyright. The contents of this proposal shall not be disclosed by the Recipient to any third party without the expressed written consent of BW Water groups. National and International patents protect core parts of our systems.

BW Water group reserves the right to alter the proposed system to incorporate technical advances realized after the date of this proposal.

BW Water's information and the recommendations presented in this proposal (hereinafter referred to as "Confidential Information") are the work product of BW Water Group and are provided solely with the understanding and agreement that the information contained in this document is submitted for evaluation only. The recipient agrees not to reveal its contents except to those in Recipient's organization necessary for evaluation and copies of this document may not be distributed outside of the Recipient organization.

By receiving and accepting this proposal, the recipient agrees to the stated terms and conditions.

Technical Proposal

Contents

EXECUTIVE SUMMARY	5
1. TECNICAL OFFER – COVER LETTER	6
DOCUMENT No. 1 – TECHNICAL DESCRIPTION	7
2. PLANT DESCRIPTION.....	8
2.1 PROJECT & SYSTEM OVERVIEW	8
2.1.1 Process Description.....	9
2.2 DESIGN BASIS	10
❖ System Recovery	10
❖ System Production Water Volume	10
❖ Feed Water Quality And quantity	11
❖ Product Water Quality	11
❖ Seismic Conditions	11
❖ Others conditions, requirements, and/or assumptions	11
3. PERFORMANCE GUARANTEES	12
3.1 PERFORMANCE PARAMETERS	12
❖ Nominal Production (permeate) Capacity	12
❖ Product Water Quality	12
❖ Specific Energy Consumption (SEC)	12
3.2 PERFORMANCE TESTS	13
❖ Nominal Permeate (Product) Capacity Test.....	13
❖ Product Water Quality Test.....	13
❖ SEC (Specific Energy Consumption) Test	13
4. SCOPE OF SUPPLY AND WORK	15
4.1 Equipment List & Technical Details	16
5. CLARIFICATIONA & DEVIATION LIST	21
6. DRAWINGS	23
6.1 3D MODEL OF CONTAINERIZED SYSTEM	23
6.2 GA / LAYOUT	25

Technical Proposal

6.3 PROCESS FLOW DIAGRAM (PFD)	27
6.4 PIPING AND INSTRUMENTATION DRAWINGS (P&ID)	29
7. RO MEMBRANE PROJECTIONS	34
8. PROTOCOLS	71
9. CONTROL PHILOSOPHY	72
10. HARDWARE AND SOFTWARE LICENSES LIST	87
11. PROJECT SCHEDULE	88
11.1 ENGINEERING SUBMITTALS & SCHEDULE	89
12. INSPECTION AND TESTING PLAN (ITP)	95
13. JUSTIFICATION OF MECHANICAL COUPLINGS/JOINTS (VICTAULIC TYPE) IN HIGH PRESSURE	103
14. MANDATORY/CRITICAL SPARE PARTS LIST FOR COMMISSIONING AND START-UP PHASES	106
15. RECOMMENDED TWO-YEAR SPARE PARTS LIST	107
16. CHEMICALS CONSUMPTION	109
17. POWER & LOAD CONSUMPTION LISTS	110
DOCUMENT No. 2 – COMMISSIONING & START-UP PLAN	112
18. COMMISSIONING AND START-UP ACTIVITIES EXECUTION PLAN	113
18.1 KEY PERSONNEL	113
18.2 PRELIMINARY ON SITE FIELD SERVICE PROGRAM	117
18.3 RESOURCES AND ASSIGNED SPECIALISTS TO BE USED	118
❖ Resources and Assigned Specialists Plan (Preliminary-Sample Plan)	118
DOCUMENT No. 3 – GUARANTEED SEC VALUE	120
19. SIGNED DECLARATION OF THE SPECIFIC ENERGY CONSUMPTION (SEC) GUARANTEED VALUE	121

Technical Proposal

EXECUTIVE SUMMARY

Technical Proposal

1. TECNICAL OFFER – COVER LETTER

Following the approval of our company, BW Water Americas Inc. (BWVA), as a qualified bidder for the public bid identified as “Proceso de Licitación 12803 – SUMINISTRO MÓDULO RO SEGUNDA ETAPA PARA SALMUERA - PLANTA DESALADORA TALTAL (ex 12783)”, we are pleased to submit our Technical Proposal for the supply of an Ultra-High-Pressure Reverse Osmosis (UHPRO) system with ancillaries for Aguas Antofagasta SA (ADASA) as part of the Brine Recovery Project at the Taltal Seawater Desalination Treatment Plant in Chile.

This proposal reflects our commitment to becoming the preferred/selected vendor for the required system package and has been organized in accordance with the instructions outlined in the document “Bases Administrativas Especiales” (BAE), Sections 20 and 20.1.

As part of our evaluation of the bidding documentation, we are proposing a containerized UHPRO system designed to treat 1,176 m³/day of brine from the existing Taltal desalination plant and produce desalinated product water that meets the Client’s project requirements and its goals of increase the production capacity of potable water for human consumption. without increasing the amount of raw water that must be extracted from the existing wells

With over 35 years of global experience in the water and wastewater industry, BWVA offers a robust, reliable, and efficient RO system tailored to meet ADASA’s water quality and production needs. Our proposed solution, which we refer to as **Brine Positive**, utilizes components from internationally recognized manufacturers and complies with the technical specifications and design drawings provided by the Client.

Our scope of supply includes a standalone, containerized second-stage UHPRO system capable of producing 21 m³/hour at a 42.85% recovery rate. The system also includes all necessary ancillary components to ensure a comprehensive and integrated solution. BWVA brings proven engineering expertise, in-house fabrication capabilities, and a global network of qualified service providers to deliver a high-performance system aligned with the Client’s goals.

We look forward to the opportunity to collaborate with ADASA and support the successful delivery of this important infrastructure project. Should you require any further information or clarification, we are ready to assist.

Best regards,

BW Water Americas Inc.

Technical Proposal

DOCUMENT No. 1 – TECHNICAL DESCRIPTION

Technical Proposal

BW Water Americas Inc. (BWWA) has prepared **Document 1 – Technical Description** in accordance with the bidding instructions outlined in the *Bases Administrativas Especiales (BAE)*, Section 20.1 – Document 1. This document also considers the list of required documentation described in *Technical Specification No. P22-ET-09-000-001-0*, Section 6.

Please note that the order in which the information is presented herein does not necessarily follow the sequence outlined in Section 6 of the Technical Specifications. However, all required elements have been addressed. Additionally, where deemed necessary and applicable, we have included supplementary information to provide a comprehensive technical offer for our proposed UHPRO system.

2. PLANT DESCRIPTION

(Ref.: P22-ET-09-000-001-0, Section 6 -Item “Descripción de la Planta”)

2.1 PROJECT & SYSTEM OVERVIEW

ADASA, the entity responsible for operating sanitary concessions for potable water production and distribution in Antofagasta, is undertaking a strategic expansion of its desalination capabilities. This project involves the supply and installation of a modular second-stage Reverse Osmosis (RO) system at the Taltal Desalination Plant. The primary objective is to increase the production of desalinated water during peak summer demand periods—without increasing the volume of raw water extracted from existing wells.

To achieve this, the required system should treat the brine (wastewater) discharged from the existing RO unit/module No3, recovering and producing a minimum of **20 m³/hour** of additional permeate. The system may accommodate up to **5% higher flow rates**, if bidders offer standardized modular solutions capable of handling larger capacities in compliance with project specifications.

BWWA proposes its **Brine Positive Plant Solution**, a brine treatment system featuring an Ultra High-Pressure Reverse Osmosis (UHPRO) unit and associated components. This advanced system is designed to maximize water recovery from existing infrastructure while maintaining high energy efficiency and operational reliability.

Our system is designed to treat **49 m³/hour (~1,176 m³/day) of brine (the feed) and produce 21 m³/hour (~504 m³/day) of permeate**. It will include:

- One (1) containerized UHPRO unit consists of Cartridge Filter, 2-stage UHPRO train, High Pressure RO Pump Energy Recovery Devices (Turbo type)
- A Lot of Ancillary equipment located outside the container, including Dosing System for high-salinity feed protection and CIP system.

Technical Proposal

2.1.1 Process Description

Brine from the existing RO module No3 at the Taltal Desalination Plant is fed into the proposed system at a flow rate of 49 m³/hour via a Brine Feed Pump (by others). The Ultra High-Pressure Reverse Osmosis (UHPRO) system operates at a recovery rate of 42.85%, producing high-quality permeate with a salinity of less than 500 ppm.

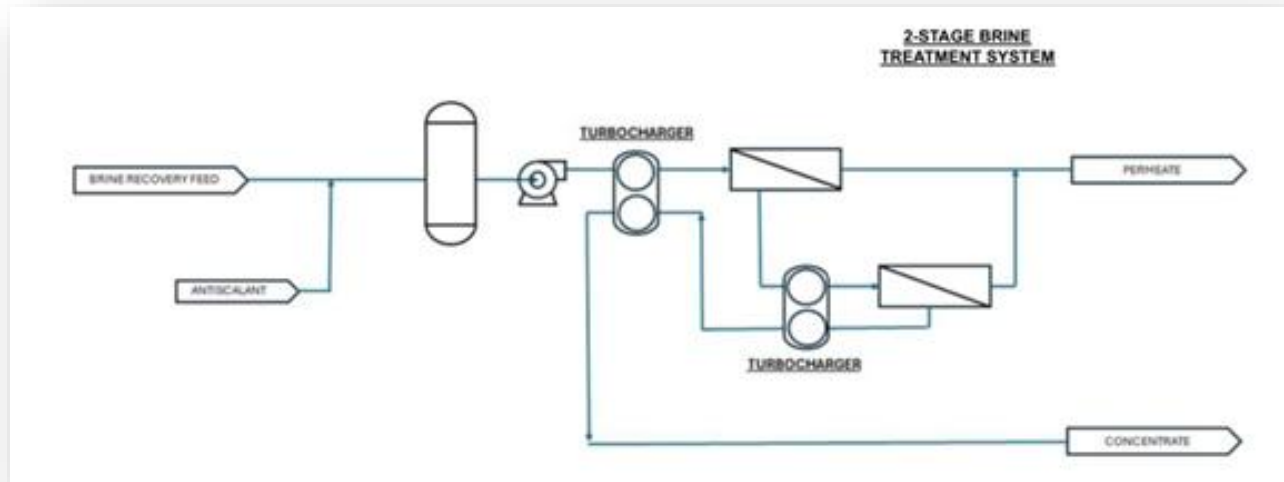
The process flow includes the following key stages:

1. **Brine Feed Transfer:** Brine supplied from the existing desalination plant (by others) is first treated with an anti-scalant to mitigate scaling risks due to high salinity. The treated feed passes through a cartridge filter before being delivered to the high-pressure RO feed pump.
2. **High-Pressure Pumping:** The filtered brine is pressurized by a high-pressure pump and directed into the first stage of the UHPRO system
3. **Feed Turbo Exchange:** An energy recovery device (Feed Turbo) boosts the pressure of the feed stream, optimizing energy efficiency and ensuring the required pressure for UHPRO operation is achieved.
4. Pressure is boosted using the Feed Turbo to meet the required UHPRO feed pressure.
5. **Interstage Turbo Exchange:** Concentrate from the first stage is further pressurized using an Interstage Turbo, which transfers energy from the second-stage concentrate to the first-stage concentrate, enabling efficient feed into the second UHPRO stage
6. **Permeate (Water Product) Transfer:** The final permeate produced by the system is dosed with anti-scalant and routed to the existing post-treatment system (provided by others) for final conditioning and distribution
7. **Brine of UHPRO system:** final Concentrate/brined produced by the system

This modular and energy-efficient solution enhances water recovery from existing infrastructure, contributing to sustainable water resource management in the region.

Here is a picture of the 2-stage concept mentioned above. Please refer to the preliminary GA/layout, PDF, P&ID and 3D drawings attached for additional details.

Technical Proposal



2.2 DESIGN BASIS

The design of the proposed Brine Treatment System is based on the technical specifications package (Tender “Licitation” No. 12803) prepared and provided by the client, ADASA, via its online portal on June 11, 2025. This package was supplemented by seven (7) formal clarifications received via email on the following dates:

- Clarification No. 01 – June 19, 2025
- Clarification No. 02 – July 1, 2025
- Clarification No. 03 – July 7, 2025
- Clarification No. 04 – July 8, 2025
- Clarification No. 05 – July 7, 2025
- Clarification No. 06 – July 28, 2025
- Clarification No. 07 – August 1, 2025

The main design conditions, requirements, and assumptions are outlined below:

❖ System Recovery

The system will have a minimum recovery of 42.85%

❖ System Production Water Volume

The system will produce 21 m³/hour of permeate

Technical Proposal

❖ Feed Water Quality And quantity

As provided by client.

Parameter	Unit	Value
Total Dissolved Solids	mg/L	43,000 - 53,300
Total Suspended Solids	mg/L	0.4 - 1
Turbidity	NTU	0.4 - 1
Temperature	°C	19 - 24

❖ Product Water Quality

Parameter	Unit	Value
Total Dissolved Solids	mg/L	< 500
Chlorides	mg/L	< 400

❖ Seismic Conditions

As applicable, the components and the system enclosure will be designed and built to continuously withstand the maximum load that may occur under all operating conditions, including earthquakes. The module structures (frame and container) and the anchors for the main internal equipment must be designed and built to withstand earthquakes in accordance with Chilean Seismic Standard NCh 2369, for Zone 3.

❖ Others conditions, requirements, and/or assumptions

The Client shall provide and guarantee the following:

- **Brine Feed Pressure** - a minimum of 45 psi (3 bar) is required
- **Feed Water Flow** - 1,176m³/day of brine

The proposed design and equipment selection is based solely upon design conditions and requirements and assumptions described thought this proposal as well as data provided by ADASA. BW Water Americas shall not be responsible for system performance issues, design and equipment changes caused by deviations to those mentioned design conditions, requirements, and/or assumptions. Performance Guarantee will be provided based upon the water quality data provided by Client, ADASA.

Technical Proposal

3. PERFORMANCE GUARANTEES

(Ref.: P22-ET-09-000-001-0, Section 6 -Item “Garantías”, (Section 10)

In accordance with Technical Specifications (P22-ET-09-000-001-00, Section 10), the Bidder (Supplier) shall guarantee the following performance parameters and demonstrate compliance through on-site performance testing, as outlined below:

3.1 PERFORMANCE PARAMETERS

❖ Nominal Production (permeate) Capacity

The **guaranteed nominal minimal capacity** for the Augas Antofagasta UHPRO Plant is **20 m³/hr (480³/day)**. ADASA (the client), will accept up to maximum permeate flow of up 21m³/hr (504 m³/day). The production is based on a feed water capacity of 1,176 m3/h (provided by client)

❖ Product Water Quality

The UHPRO Plant will produce water that meets the **minimum quality conditions** defined below, contingent upon the feed water quantity and quality provided by ADASA meeting the requirements stated in Section 2.2 of this proposal:

Parameter	Unit	Value
Total Dissolved Solids	mg/L	< 500
Chlorides	mg/L	< 400

❖ Specific Energy Consumption (SEC)

SEC is defined as the **power consumption of the RO Unit - specifically the RO high-pressure pump only** -during the performance testing period, divided by the actual volume of product water (21 m³/hr) at a feed TDS of 53,000 ppm.

- **Guaranteed SEC Value:** will be < 5.0 kWh/m³ at a feed TDS of 53,000 ppm. Please refer to Section 19 for the signed Declaration on our proposed Guaranteed SEC Value
- **Extended SEC Value** which includes chemical dosing pump, CIP/flushing pump, CIP heater, and RO PLC power, but excluding container AC and lighting): **5.42 kWh/m³ ± 5%**

Please refer to Section 3.2 below, Section19 guaranteed SEC value, and Section 17 - Power Consumption Calculations of this proposal

Technical Proposal

3.2 PERFORMANCE TESTS

The plant shall operate continuously for 2 consecutive days at 100% capacity. This test will validate the guaranteed parameters for:

- **Nominal Capacity**
- **SEC**
- **Product Water Quality**
- ❖ **Nominal Permeate (Product) Capacity Test**

The permeate **capacity of 21 m³/hr** will be delivered to the Client's plant for reuse and **measured via the permeate flowmeter** to confirm compliance.

- ❖ **Product Water Quality Test**

Daily samples will be collected from the permeate side during the 2-day test period.

- **Chloride and TDS** will be measured using an **on-site conductivity analyzer instrument** and **laboratory analysis**. See table above for minimum water quality conditions (Section 3.1)
- Laboratory testing will be conducted by **ADASA**, with sampling performed by **BWWA** (subject to price adder).

Note: For Product Water Quality, BWWA warrants that the products shall perform in accordance with the technical specifications set forth in minimum water quality conditions (Section 3.1) for the period of twelve (12) months commencing from successful completion of the membrane element performance. These twelve (12) months limited performance warranty is specified and confirmed by membrane manufacturer.

- ❖ **SEC (Specific Energy Consumption) Test**

Total energy consumption during the test period will be recorded via the UHPRO Plant Power Panel Meter.

- The **guaranteed SEC value** shall remain below **5.0 kWh/m³** at a TDS of 53,000 ppm. Our guaranteed SEC value calculated as consumption of RO HP Pump only divided by the Permeate capacity, at the Max TDS level of 53,000 ppm will **be 4.45 kWh/m³**. Please refer to Section 19 –Declaration of Guaranteed SEC Value” of this proposal. **This SEC Value is one to be considered for the SEC performance guaranteed and the commercial evaluation of the offer.**

Technical Proposal

- The **extended SEC**, including auxiliary systems (excluding container AC and lighting), must not exceed **5.42 kWh/m³ ± 5%** at a TDS of 53,000 ppm. **This is not part of the performance guarantee.** Please refer to Section 17-Power Consumption Calculations of this proposal.

Technical Proposal

4. SCOPE OF SUPPLY AND WORK

BWWA will be responsible for the engineering design, procurement/fabrication, and supply of a modular second-stage Reverse Osmosis (RO) system, along with applicable ancillary equipment, to be installed at ADASA's Taltal Desalination Plant (installation by others). The scope also includes related services such as installation supervision, commissioning and start-up, and operator training.

The table below summarizes the major items / equipment included in our proposed solution.

A more comprehensive list with some additional details is presented in the attached equipment lists (section 4.1.1 below)

Quantity	Mayor Items/ Equipment
UHPRO Module -Inside of the container	
1 x 100%	RO Feed Cartridge Filter package – Fiberglass housing with 15 polypropylene elements (2.5" x 40") per filter. Installed inside a 40 ft container with piping, valves, and instrumentation.
1 x 100%	RO High-Pressure Pump with VFD package – Super Duplex casing and impeller, 49 m ³ /h (215.6 GPM) @ 723 psig (49.14 barg), 115.3 hp (86 kW) motor. Installed inside a 40 ft container with piping, valves, and instrumentation.
1 x 100%	Double Stage Ultra High-Pressure RO (UHPRO) Train with ERD package – 21 m ³ /h (92.4 GPM) permeate capacity per train, designed for 42.85% recovery. Configuration: 6:4 array, 7-element vessels, 70 TFC RO membranes. Includes one Super Duplex Bi-Turbo Energy Recovery package (1 Feed Turbo and 1 Interstage Turbo), SDSS high-pressure piping, PVC low-pressure permeate piping, and all necessary valving/instrumentation tubed and wired to a junction box. Installed inside a 40 ft container.
Ancillary Equipment of the UHPRO Module - Out of the container	
1 x 100%	CIP / Flushing Tank package – HDPE material, 5.1 m ³ capacity. Includes CIP heater and is installed with piping, valves, and instrumentation. Located outside the 40 ft container.
1 x 100%	CIP / Flushing Pump package – SS316 casing and impeller, 57 m ³ /h (250 GPM) @ 61 psig (4.1 barg), 15 hp (11 kW) motor. Installed with piping, valves, and instrumentation. Located outside the 40 ft container.
1 x 100%	CIP Cartridge Filter package – Fiberglass housing with 15 polypropylene elements (2.5" x 40") per filter. Installed with piping, valves, and instrumentation. Located outside the 40 ft container.

Technical Proposal

Quantity	Mayor Items/ Equipment
1 x 100%	Anti-Scalant Injection System package – Includes 1 HDPE drum, 1 pump skid, and plastic piping. Installed with piping, valves, and instrumentation. Located outside the 40 ft container.
1 Lot	Local Control Panel with HMI package – Installed outside the 40 ft container.

Exclusions:

The following items are excluded from the scope of supply and services for the proposed Brine Treatment System:

- Motor Control System
- Civil and Foundation Design
- Civil Works
- Installation of Supplied Equipment/System and direct labor during commissioning and start-up. BWWA will provide supervision for these on-site services, as detailed in Section 18 of this proposal.

Also, please refer to Section 5 of this proposal

4.1 Equipment List & Technical Details

(Ref.: P22-ET-09-000-001-0, Section 6 -Item “Especificación de equipos electromecánicos con marcas y modelos” | -Item “Especificación de equipos eléctricos con marcas y modelos” | -Item “Especificación de instrumentos y PLC con marcas y modelos”)

The lists provided herein are preliminary. The complete and final equipment/component lists, as applicable, will be developed and submitted during the execution phase of the project. These will be included as part of the engineering submittals and will reflect the outcomes of technical discussions and mutual agreement on the final scope of work. It is understood that modifications may occur as the project progresses and requirements are refined.

- ❖ Electromechanical equipment List
- ❖ PLC & Instruments List

		Mechanical Equipment List						
		CUSTOMER:	Aguas Antofagasta				REV.:	0
		LOCATION:	Chile				DATE:	July 22, 2025
		PROJECT No.:	Proposal 20.24.6501 -ADASA Brine Recovery Project -UHPRO System				ORIG. BY:	
Item	Description /Service	Qty	Capacity	Material	Dimensions	Details	Manufacturers	Notes
1	Static Mixer	1 x 100%	49 m3/hr	FRP	4" (D) x 22" (H)	One Ports (1" CL150 RFSO Flanged) Connection : (CL 150 RFSO Flanged) Triple Action Mixing Elements	KOMAX or Equal	
2	RO Cartridge Filter	1 x 100%	49 m3/hr	FRP Housing Polypropylene filter	Shell(FRP) : 12 in (D) x 55in (H) Filter : 2 1/2" OD x 40"	# of filters in housing : 12	Filtrek or Equal	
3	RO HP Feed Pump	1 x 100%	51.45 m3/hr @ 49.14 Barg	SDSS Shaft/Shell	123 in (L) x 34 in (W) x 28.9 in (H)	Motor Rating : 86kW IP55/Class B (Protection) VFD (Strater Type)	Fedco or Equal	
4	Feed Turbo	1 x 100%	Feed : 49 m3/hr (Membrane Pressure : 57.0 barg) Brine : 27.9 m3/hr (Outlet Pressure :1 barg)	SDSS 2507	11.51 in (L) x 11(W) x 9 in (H)	Model : HPB - 60	Fedco or Equal	
5	Interstage Turbo	1 x 100%	Feed : 35.4 m3/hr (Membrane Pressure : 71.5 barg) Brine : 27.9 m3/hr (Outlet Pressure :1 barg)	SDSS 2507	11.51 in (L) x 11(W) x 9 in (H)	Model : HPB - 60	Fedco or Equal	
6	Doble stage Brine Treatment RO Unit	1 x 100%						See item 6.1 and 6.2
6.1	RO Memebrane	70	Permeate Flow : 21 m3/hr Recovery : 42.85%	Composite Polyamide	8" (D) x 40" (L)	Membrane : LG SW 400 R G2 UHP; 42 Membranes for 1st stage and 28 membrend for 2nd stage	LG (or Dupont)	
6.2	RO Pressure Vessesl (PV)/ Tubes	10	Array: (10PVs) 6:4 , 7 membrane elements per PV	FRP/SDSS	13.92 " (D) x 343.2 " (L)	To be provided later for details.	Protec or Equal	
7	CIP/Flushing Tank	1 x 100%	5.1 m3 (1350 GAL)	HDPE	71in (D) x 88in (H)		NTO Tank or Equal	
8	CIP Heater	1 x 100%	20 kW	316 SS/316L SS Flange	3" Flanged 7 1/2 in (D) x 47 1/2 in (L) .	150 lb	CHROMALOX or Equal	
9	CIP/Flushing Pumps	1 x 100%	54.5 m3/hr @ 4.13 barg	316SS Shaft/Shell	30 in (L) x 13 in (W) x 14.4 in (H)	150 lb ANSI Flange Motor : 15kW (480V/50Hz/3Ph) IP55/Class B VFD (Starter Type)	Groundfos or Equal	
10	CIP/Flushing Cartridge Filter	1 x 100%	54.5 m3/hr	SS316 Housing Polypropylene filter	Shell(FRP) : 14.50 in (D) x 55 in (H) Filter : 2 1/2" OD x 40"	# of filters in housing : 19	Filtrek or Equal	

		Mechanical Equipment List						
		CUSTOMER:	Aguas Antofagasta				REV.:	0
		LOCATION:	Chile				DATE:	July 22, 2025
		PROJECT No.:	Proposal 20.24.6501 -ADASA Brine Recovery Project -UHPRO System				ORIG. BY:	
Item	Description /Service	Qty	Capacity	Material	Dimensions	Details	Manufacturers	Notes
11	Antiscalant Dosing Pumps (Skid)	2 x 100%	1 LPH @ 4 barg	PVC wetted parts and piping Skid : Duplex	9.98 in (L) x 6.71 in (W) x 12.25 in (H)	Solenoid, Positive Displacement Internal Pressure Relief Valve Control : Automatic Seal Material : Standard	Groundfos or Equal	
12	Antiscalant Dosing Tank	1 x 100%	65 gal	HDPE	23" (D) x 26"(H) x 43"(L)		Norwesco or Equal	

		PLC and Intruments List					
		CUSTOMER:	Aguas Antofagasta			REV.:	0
		LOCATION:	Chile			DATE:	25-Jul-25
		PROJECT No.:	Proposal 20.24.6501 -ADASA Brine Recovery Project -UHPRO System			ORIG. BY:	Nicholaus Huta
Instrument List							
Item	Instrument Type / Description	Media	Manufacturer	Location	Qty	Notes	
1	Analytical Transmitter - ORP	Seawater Brine	EMERSON / ABB / EQUAL	RO CF DISCHARGE	1		
2	Analytical Transmitter - CHLORINE	Seawater Brine	EMERSON / ABB / EQUAL	RO CF DISCHARGE	1		
3	Pressure Transmitter	Seawater Brine	FOXBORO / EQUAL	RO HP PUMP SUCTION	1		
4	Pressure Transmitter	Seawater Brine	FOXBORO / EQUAL	RO HP PUMP DISCHARGE	1		
5	Pressure Transmitter	Seawater Brine	FOXBORO / EQUAL	RO STAGE 1 FEED	1		
6	Pressure Transmitter	Seawater Brine	FOXBORO / EQUAL	RO STAGE 1 REJECT	1		
7	Pressure Transmitter	Seawater Brine	FOXBORO / EQUAL	RO STAGE 2 FEED	1		
8	Pressure Transmitter	Seawater Brine	FOXBORO / EQUAL	RO STAGE 2 REJECT	1		
9	Pressure Transmitter	Seawater Brine	FOXBORO / EQUAL	INTERSTAGE TURBO TO FEED TURBO	1		
10	Pressure Transmitter	Seawater Brine	FOXBORO / EQUAL	RO TRAIN COMBINED PERMEATE	1		
11	Pressure Transmitter	Seawater Brine	FOXBORO / EQUAL	RO TRAIN REJECT	1		
12	Pressure Gauge	Seawater Brine	ASHCROFT / WIKA / EQUAL	RO HP PUMP DISCHARGE	1		
13	Pressure Gauge	Seawater Brine	ASHCROFT / WIKA / EQUAL	RO TRAIN REJECT	1		
14	Flow Transmitter	Seawater Brine	ROSEMOUNT / ABB / EQUAL	RO TRAIN COMBINED PERMEATE	1		
15	Flow Transmitter	Seawater Brine	ROSEMOUNT / ABB / EQUAL	RO TRAIN STAGE 2 PERMEATE	1		
16	Flow Transmitter	Seawater Brine	ROSEMOUNT / ABB / EQUAL	RO TRAIN REJECT	1		
17	Analytical Transmitter - CONDUCTIVITY	Seawater Brine	EMERSON / ABB / EQUAL	RO TRAIN COMBINED PERMEATE	1		
18	Motorized Valve	Seawater Brine	EMERSON / EQUAL	RO TRAIN PERMEATE OUT	1		
19	Motorized Valve	Seawater Brine	EMERSON / EQUAL	OFF-SPEC PERMEATE DUMP	1		
20	Motorized Valve	Seawater Brine	EMERSON / EQUAL	CIP TO RO STAGE 1	1		
21	Motorized Valve	Seawater Brine	EMERSON / EQUAL	CIP TO RO STAGE 2	1		
22	Motorized Valve	Seawater Brine	EMERSON / EQUAL	FEED TURBO LP BYPASS	1		
23	Motorized Valve	Seawater Brine	EMERSON / EQUAL	INTERSTAGE TURBO LP BYPASS	1		
24	Motorized Valve	Seawater Brine	EMERSON / EQUAL	RO CIP PERM RETURN	1		
25	Motorized Valve	Seawater Brine	EMERSON / EQUAL	RO STAGE 1 CIP REJECT	1		
26	Motorized Valve	Seawater Brine	EMERSON / EQUAL	RO STAGE 2 CIP REJECT	1		
27	Temperature Transmitter	Seawater Brine	ROSEMOUNT / SIEMENS / EQUAL	CIP TANK	1		
28	Level Gauge	Seawater Brine		CIP TANK	1		
29	Level Transmitter	Seawater Brine	IFM / VEGA / EQUAL	CIP TANK	1		
30	Pressure Gauge	Seawater Brine	ASHCROFT / WIKA / EQUAL	CIP PUMP DISCHARGE	1		
31	Pressure Gauge	Seawater Brine	ASHCROFT / WIKA / EQUAL	CIP CF INLET	1		
32	Pressure Gauge	Seawater Brine	ASHCROFT / WIKA / EQUAL	CIP CF OUTLET	1		
33	Analyzer Indicator - pH	Seawater Brine	EMERSON / ABB / EQUAL	CIP PUMP DISCHARGE	1		
34	Flow Transmitter	Seawater Brine	ROSEMOUNT / ABB / EQUAL	CIP TO RO TRAIN	1		
35	Motorized Valve	Seawater Brine	EMERSON / EQUAL	ANTISCALANT TO DOSING TANK	1		
36	Level Gauge	Seawater Brine		ANTISCALANT TANK	1		
37	Level Transmitter	Seawater Brine	IFM / VEGA / EQUAL	ANTISCALANT TANK	1		
38	Pressure Gauge	Seawater Brine	ASHCROFT / WIKA / EQUAL	ANTISCALANT PUMP DISCHARGE	1		

		PLC and Intruments List					
		CUSTOMER: Aguas Antofagasta				REV.:	0
		LOCATION: Chile				DATE:	25-Jul-25
		PROJECT No.: Proposal 20.24.6501 -ADASA Brine Recovery Project -UHPRO System				ORIG. BY:	Nicholaus Huta
PLC Major Parts BOM							
Item	Model Number /	Description	Type	Manufacturer	Location	Qty	Notes
1	1769-L18ER-BB1B		CPU	Allen Bradley	PLC PANEL	1	
2	1734-IB8		DIGITAL INPUT	Allen Bradley	PLC PANEL	A/R	
3	1734-OB8		DIGITAL OUTPUT	Allen Bradley	PLC PANEL	A/R	
4	1734-IE8C		ANALOG INPUT	Allen Bradley	PLC PANEL	A/R	
5	1734-OE4C		ANALOG OUTPUT	Allen Bradley	PLC PANEL	A/R	
6	Panelview Plus 7		HMI	Allen Bradley	PLC PANEL	1	

Technical Proposal

5. CLARIFICATIONA & DEVIATION LIST

No.	Subjects	Contents	Reference
1	Technical Bid Submittal Document	"Protocols to be used" will be submitted as detail in execution stage, BWWA will choose communication as Ethernet/IP	P22 -ET-09-000-001 Page 26, 6. DOCUMENTATION TO BE PRESENTED IN THE PROPOSAL
2	Guarantee	Guaranteed specific energy consumption, <5kWh/m ³ , should be confined into RO HP pump energy consumption considering maximum TDS, 53,000ppm.	P22 -ET-09-000-001 Page 26, 10. GUARANTEE
3	Guarantee	Product Water Quality Test through Laboratory analysis shall be conducted by ADASA and sampling will be conducted by BWWA but this will be price adder	P22 -ET-09-000-001 Page 26, 10. GUARANTEE
4	Technical Bid Submittal Document	BWWA shall provide List of Hardware and Software License for HMI/PLC Programming and this will be Hardware and Software License for HMI/PLC Programming as a price adder.	P22 -ET-09-000-001 Page 26, 6. DOCUMENTATION TO BE PRESENTED IN THE PROPOSAL
5	Guarantee	The total energy consumption of the system—considering not only the HP pumps but also the chemical dosing pump, CIP/flushing pump, CIP heater, and RO PLC power—is estimated at 5.42 kWh/m³ ± 5%. This figure excludes the AC power and lighting within the container. While this value exceeds the specified guarantee of <5 kWh/m³, it represents the total power consumed by the plant, divided by the permeate capacity of 21 m³/hr. For clarification on the guaranteed SEC value applicable to the Performance Guarantee and the extended SEC value, please refer to: Section 3, and Section 17 and Section 19	P22 -ET-09-000-001 Page 26, 10. GUARANTEE

Technical Proposal

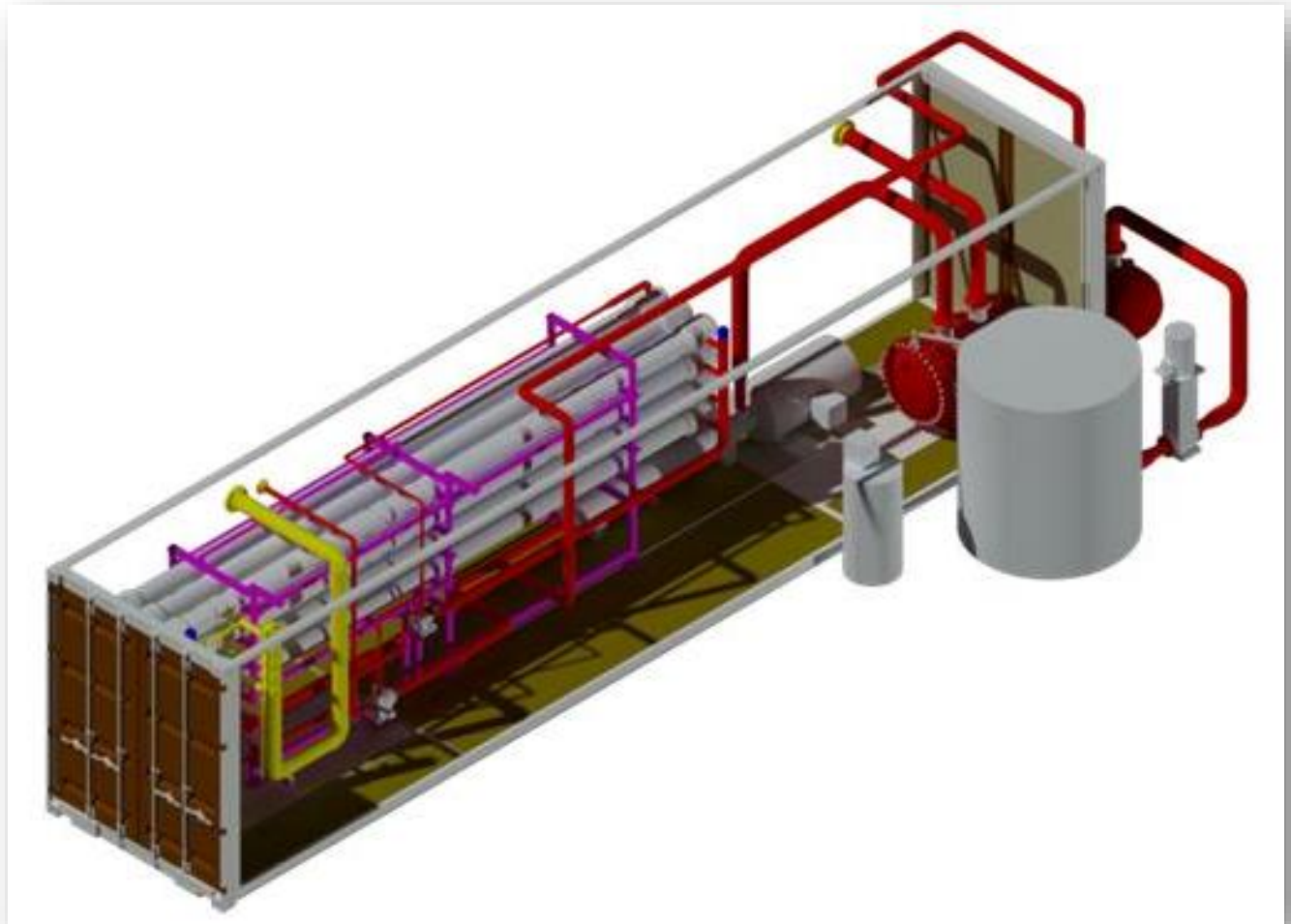
No.	Subjects	Contents	Reference
6	Submission Due date	ADASA has stated that the supplier must submit all required documentation for approval no later than 90 calendar days following the award notice. However, we are considering a maximum delivery time of 42 weeks from the date of purchase order (PO)/contract acceptance, most documentation related to the engineering phase is expected to be submitted within 15 weeks of PO/contract acceptance—not from the award notice. It is estimated that there will be a 15 to 20-day gap between the award notice and the completion of PO/contract acceptance.	P22 -ET-09-000-001 Page 26, 7. Engineering and Documentation to be developed during the charge
7	Document submittal during execution stage	Flexibility analysis of high-pressure lines is vague. BWWA can provide the class rating for the high-pressure line to show flexibility analysis.	P22 -ET-09-000-001 Page 26, 7. Engineering and Documentation to be developed during the charge

Technical Proposal

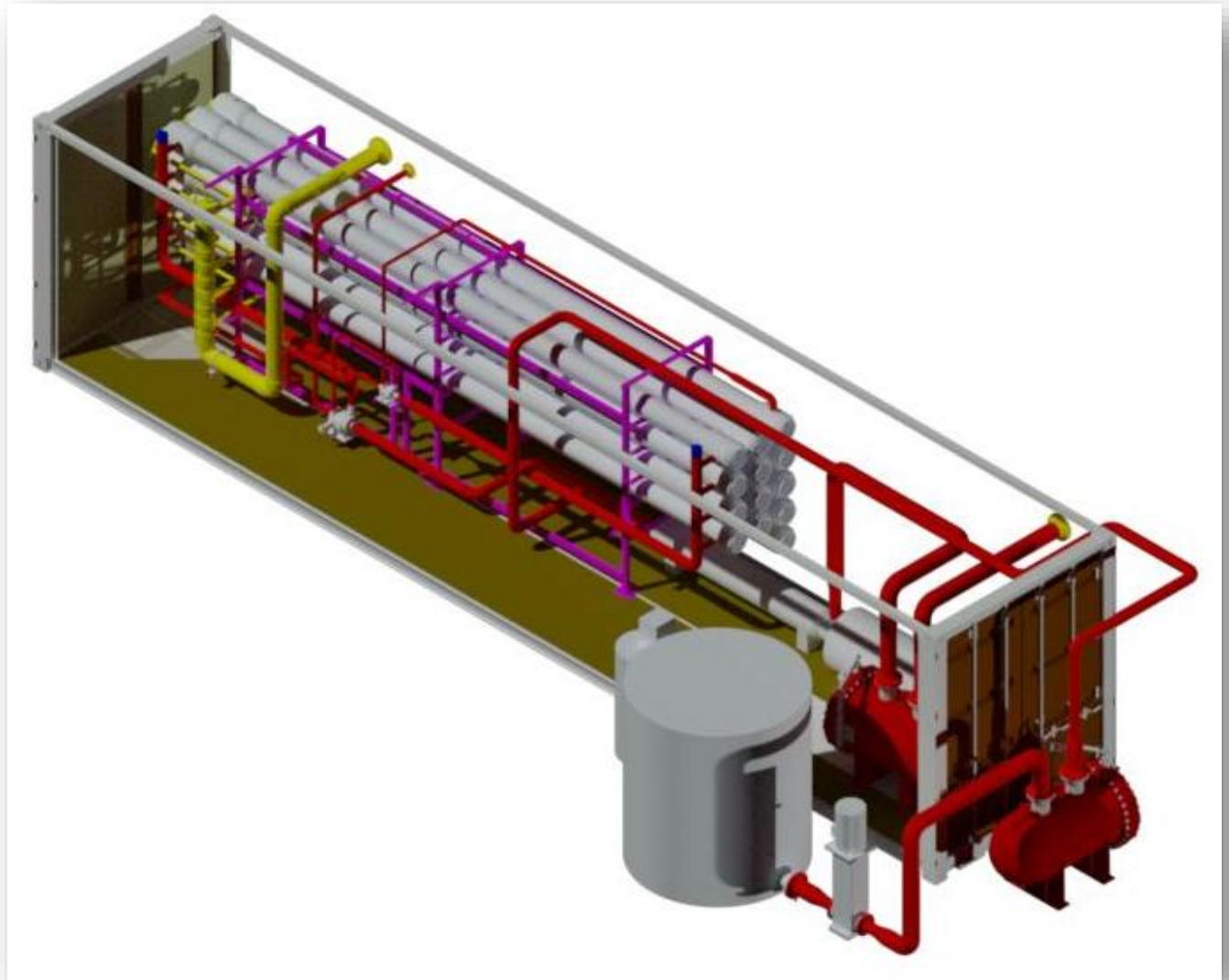
6. DRAWINGS

The following preliminary diagrams and drawings are included as part of this proposal.

6.13D MODEL OF CONTAINERIZED SYSTEM



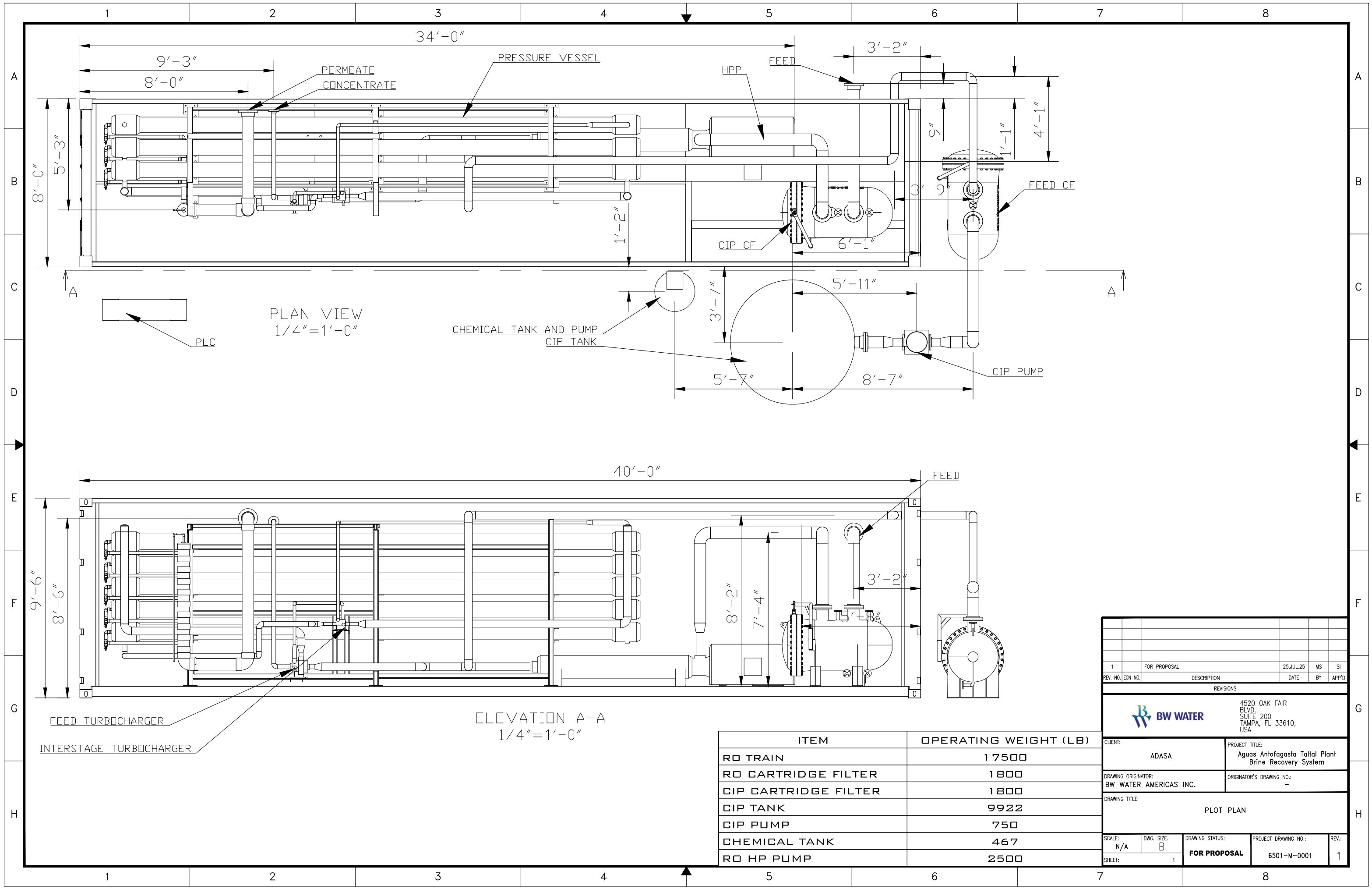
Technical Proposal



Technical Proposal

6.2 GA / LAYOUT

(Ref.: P22-ET-09-000-001-0, Section 6 -Item "Layout con las dimensiones del modulo")

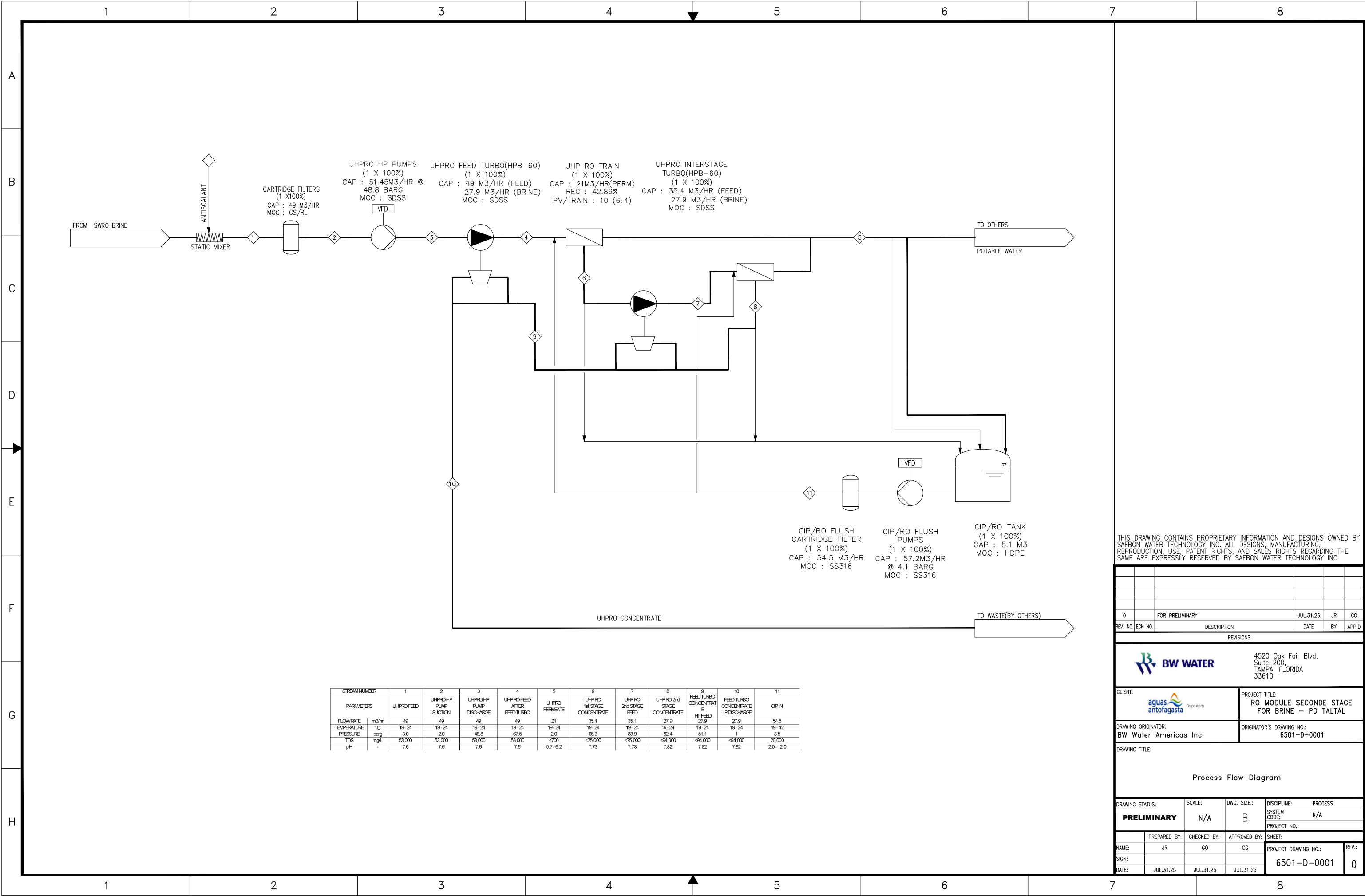


1		FOR PROPOSAL	25.JUL.25	MS	SI
REV. NO.	ECON NO.	DESCRIPTION	DATE	BY	APP'D
REVISIONS					
		4520 OAK FAIR BLVD. SUITE 200 TAMPA, FL 33610, USA			
CLIENT: ADASA		PROJECT TITLE: Aguas Antofagasta Taltal Plant Brine Recovery System			
DRAWING ORIGINATOR: BW WATER AMERICAS INC.		ORIGINATOR'S DRAWING NO.: —			
DRAWING TITLE: PLOT PLAN					
SCALE: N/A	DWG. SIZE.: B	DRAWING STATUS: FOR PROPOSAL	PROJECT DRAWING NO.: 6501-M-0001		REV.: 1
SHEET: 1					

Technical Proposal

6.3 PROCESS FLOW DIAGRAM (PFD)

(Ref.: P22-ET-09-000-001-0, Section 6 -Item “Diagrama de flujo con presiones y balances de masas”)



THIS DRAWING CONTAINS PROPRIETARY INFORMATION AND DESIGNS OWNED BY SAFBON WATER TECHNOLOGY INC. ALL DESIGNS, MANUFACTURING, REPRODUCTION, USE, PATENT RIGHTS, AND SALES RIGHTS REGARDING THE SAME ARE EXPRESSLY RESERVED BY SAFBON WATER TECHNOLOGY INC.

0		FOR PRELIMINARY	JUL.31.25	JR	GO
REV. NO.	EDN NO.	DESCRIPTION	DATE	BY	APP'D

REVISIONS



4520 Oak Fair Blvd,
Suite 200,
TAMPA, FLORIDA
33610

CLIENT:



PROJECT TITLE:
RO MODULE SECONDE STAGE
FOR BRINE – PD TALTAL

DRAWING ORIGINATOR:

BW Water Americas Inc.

ORIGINATOR'S DRAWING NO.:

6501-D-0001

DRAWING TITLE:

Process Flow Diagram

DRAWING STATUS:

PRELIMINARY

SCALE:

N/A

DWG. SIZE:

B

DISCIPLINE:

PROCESS

SYSTEM CODE:

N/A

PROJECT NO.:

NAME:

JR

CHECKED BY:

GO

APPROVED BY:

OG

SHEET:

SIGN:

JUL.31.25

JUL.31.25

JUL.31.25

PROJECT DRAWING NO.:

6501-D-0001

REV.:

0

Technical Proposal

6.4 PIPING AND INSTRUMENTATION DRAWINGS (P&ID)

(Ref.: P22-ET-09-000-001-0, Section 6 -Item "P&ID")

